

POLI666 Manual Replication - MASTER FILE

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Setup and data loading

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

set.seed(666)

library(tidyverse)
library(haven)
library(lmtest)
library(sandwich)
library(modelsummary)
library(kableExtra)
library(psych)
library(lavaan)

data <- read_dta("moscowpopsurvey.dta", encoding = "Windows-1251")
```

Cleaning data: Constructing variables

Building three time periods

```
data <- data %>%
  rename(day = DATA) %>%
  mutate(period = ifelse(day > 24 | day < 5, 1, 0),
         dec25 = ifelse(MONTH == 12 & day == 25, 1, 0),
         period1 = period - dec25,
         period2 = ifelse(day > 4 & day == 11, 1, 0),
         electday = ifelse(MONTH == 12 & day == 4, 1, 0),
         #electday = "Election Day"
         protestday = ifelse(MONTH == 12 & day == 10, 1, 0),
         #protestday = "Protest Day"
         period1min = period1 - electday,
         period2min = period2 - protestday,
         #period2min = "Post-Election"
         period3 = ifelse(day > 10 & day < 26 & MONTH == 12, 1, 0))
         #period3 = "Post-Protest"
```

Building trust measures

```
data <- data %>%
  mutate(trustarmy = case_match(q9a, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
                                .default = q9a), #Army
```

```

trustpolice = case_match(q9b, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9b), #Police
trustfsb = case_match(q9c, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9c), #FSB
trustcourt = case_match(q9d, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9d), #Court
trustmungov = case_match(q9e, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9e), #City Gov't
trustfedgov = case_match(q9f, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9f), #Fed Gov't
trustduma = case_match(q9g, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9g), #Duma
trustprocuracy = case_match(q9h, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9h), #Procuracy
trustchurch = case_match(q9i, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9i), #Church
#Church was not included in the original paper's analysis,
#but is included here for subsequent analysis/checks.
trustun = case_match(q9j, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
  .default = q9j)) #UN

#joining trust measures into gov trust index (trustgovind)

data <- data %>%
  mutate(trustgovind = (trustarmy + trustpolice + trustfsb + trustcourt +
    trustmungov + trustfedgov + trustduma +
    trustprocuracy)/8)
#Gov't Index (as built by Frye and Borisova 2019)

```

Mapping out/recoding potential confounders

```

data <- data %>%
  rename(age = q1,
    male = q2,
    wealth = q3,
    education = q5,
    permres = q59,
    nationality = q67) %>%
  mutate(male = ifelse(male == 2, 0, 1),
    wealth = case_match(wealth, 99 ~ NA, .default = wealth),
    education = case_match(education, 99 ~ NA, .default = education),
    permres = case_match(permres, 99 ~ NA, 2 ~ 0, 3 ~ 0,
      .default = permres),
    nationality = case_match(nationality, 2 ~ 0, 3 ~ 0, 4 ~ 0, 5 ~ 0,
      6 ~ 0, 7 ~ 0, 8 ~ 0, 9 ~ 0, 10 ~ 0, 11 ~ 0,
      99 ~ 0, .default = nationality),
    unemp = ifelse(q6 == 11 | q6 == 12 | q6 == 13, 1, 0),
    statesector = ifelse(q7 > 9 & q7 < 13, 1, 0),
    okrug = factor(okrug))

table(data$okrug)

```

```
##
```

```
## 1 2 3 4 5 6 7 8 9 10
## 107 146 173 200 173 232 202 168 120 29
```

Coding partisanship

```
#nesting each partisanship variable in its own mutate() function
#q62 = "Which party do you intend to vote for?"
#q64 = "Which party did you vote for in the previous election?"
data <- data %>%
  mutate(urplan = 0, urturn = 0,
         urplan = ifelse(!is.na(q62) & q62 == 6, 1, 0),
         urturn = ifelse(!is.na(q64) & q64 == 6, 1, 0),
         ur = urplan + urturn) %>% #united russia variable
  mutate(complan = 0, compturn = 0,
         complan = ifelse(!is.na(q62) & q62 == 3, 1, 0),
         compturn = ifelse(!is.na(q64) & q64 == 3, 1, 0),
         comp = complan + compturn) %>% #CPFR party (?)
  mutate(ldprplan = 0, ldprturn = 0,
         ldprplan = ifelse(!is.na(q62) & q62 == 4, 1, 0),
         ldprturn = ifelse(!is.na(q64) & q64 == 4, 1, 0),
         ldpr = ldprplan + ldprturn) %>% #LDPR party
  mutate(yabplan = 0, yabturn = 0,
         yabplan = ifelse(!is.na(q62) & q62 == 1, 1, 0),
         yabturn = ifelse(!is.na(q64) & q64 == 1, 1, 0), #Yabloko party
         yab = yabplan + yabturn) %>%
  mutate(patplan = 0, patturn = 0,
         patplan = ifelse(!is.na(q62) & q62 == 2, 1, 0),
         patturn = ifelse(!is.na(q64) & q64 == 2, 1, 0),
         pat = patplan + patturn) %>% #Unsure of party
  mutate(srplan = 0, srturn = 0,
         srplan = ifelse(!is.na(q62) & q62 == 5, 1, 0),
         srturn = ifelse(!is.na(q64) & q64 == 5, 1, 0),
         sr = srplan + srturn) %>% #Fair Russia party
  mutate(rcplan = 0, rcturn = 0,
         rcplan = ifelse(!is.na(q62) & q62 == 7, 1, 0),
         rcturn = ifelse(!is.na(q64) & q64 == 7, 1, 0),
         rc = rcplan + rcturn) %>% #Unsure of party
  mutate(dkplan = ifelse(!is.na(q62) & q62 == 99, 1, 0),
         dkturn = ifelse(!is.na(q64) & q64 == 99, 1, 0),
         dk = dkplan + dkturn) %>% #people who "don't know" (q62/q64 = 99)
  mutate(nopart = 0,
         nopart = ifelse((!is.na(q61) & q61 == 2) | (!is.na(q63) & q63 == 2) |
                        (!is.na(q62) & q62 == 0), 1, 0)) #no party

#q61 = "Do you intend to vote in the upcoming (Duma) elections?"
#q63 = "Did you vote in the previous (Duma) elections?"
```

Coding non-response variables

```
data <- data %>%
  mutate(trustgovind.nr = case_when(trustgovind >= 0.85 ~ 1, NA ~ 1, TRUE ~ 0),
         trustarmy.nr = case_match(q9a, 99 ~ 1, .default = 0),
         trustpolice.nr = case_match(q9b, 99 ~ 1, .default = 0),
```

```

trustfsb.nr = case_match(q9c, 99 ~ 1, .default = 0),
trustcourt.nr = case_match(q9d, 99 ~ 1, .default = 0),
trustmungov.nr = case_match(q9e, 99 ~ 1, .default = 0),
trustfedgov.nr = case_match(q9f, 99 ~ 1, .default = 0),
trustduma.nr = case_match(q9g, 99 ~ 1, .default = 0),
trustprocuracy.nr = case_match(q9h, 99 ~ 1, .default = 0),
trustchurch.nr = case_match(q9i, 99 ~ 1, .default = 0),
trustun.nr = case_match(q9j, 99 ~ 1, .default = 0))

```

Results

Table 1

```

#period 1 vs. period 2
p12.data <- data %>% filter(period3 != 1 & electday != 1 & protestday !=1)

#NOTE - can later pull:
#result$statistic - t-stat
#result$p.value - p-value
#result$conf.int ~ 95% CI
#result$estimate - mean(s) of the group(s)
#result$parameter - df
#result$stderr - SE

p12.army <- t.test(trustarmy ~ period1min, data = p12.data)
p12.police <- t.test(trustpolice ~ period1min, data = p12.data)
p12.fsb <- t.test(trustfsb ~ period1min, data = p12.data)
p12.court <- t.test(trustcourt ~ period1min, data = p12.data)
p12.mungov <- t.test(trustmungov ~ period1min, data = p12.data)
p12.fedgov <- t.test(trustfedgov ~ period1min, data = p12.data)
p12.duma <- t.test(trustduma ~ period1min, data = p12.data)
p12.procuracy <- t.test(trustprocuracy ~ period1min, data = p12.data)
p12.un <- t.test(trustun ~ period1min, data = p12.data)
p12.trustgovind <- t.test(trustgovind ~ period1min, data = p12.data)

#period 1 vs. period 3
p13.data <- data %>% filter(period2min != 1 & electday != 1 & protestday !=1)

p13.army <- t.test(trustarmy ~ period1min, data = p13.data)
p13.police <- t.test(trustpolice ~ period1min, data = p13.data)
p13.fsb <- t.test(trustfsb ~ period1min, data = p13.data)
p13.court <- t.test(trustcourt ~ period1min, data = p13.data)
p13.mungov <- t.test(trustmungov ~ period1min, data = p13.data)
p13.fedgov <- t.test(trustfedgov ~ period1min, data = p13.data)
p13.duma <- t.test(trustduma ~ period1min, data = p13.data)
p13.procuracy <- t.test(trustprocuracy ~ period1min, data = p13.data)
p13.un <- t.test(trustun ~ period1min, data = p13.data)
p13.trustgovind <- t.test(trustgovind ~ period1min, data = p13.data)

#period 2 vs. period 3
p23.data <- data %>% filter(period1min != 1 & electday != 1 & protestday !=1)

p23.army <- t.test(trustarmy ~ period2min, data = p23.data)
p23.police <- t.test(trustpolice ~ period2min, data = p23.data)

```

```

p23.fsb <- t.test(trustfsb ~ period2min, data = p23.data)
p23.court <- t.test(trustcourt ~ period2min, data = p23.data)
p23.mungov <- t.test(trustmungov ~ period2min, data = p23.data)
p23.fedgov <- t.test(trustfedgov ~ period2min, data = p23.data)
p23.duma <- t.test(trustduma ~ period2min, data = p23.data)
p23.procuracy <- t.test(trustprocuracy ~ period2min, data = p23.data)
p23.un <- t.test(trustun ~ period2min, data = p23.data)
p23.trustgovind <- t.test(trustgovind ~ period2min, data = p23.data)

```

Table 2

```

#running all regression separately

lm.govindex <- lm(trustgovind ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.army <- lm(trustarmy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.police <- lm(trustpolice ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.fsb <- lm(trustfsb ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.court <- lm(trustcourt ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.mungov <- lm(trustmungov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.fedgov <- lm(trustfedgov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.duma <- lm(trustduma ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.procuracy <- lm(trustprocuracy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.un <- lm(trustun ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

#reporting in one table
modelsummary(list("Gov Index" = lm.govindex, "Army" = lm.army, "Police" = lm.police,
  "FSB" = lm.fsb, "Court" = lm.court, "Municipal Gov" = lm.mungov,
  "Federal Gov" = lm.fedgov, "Duma" = lm.duma,
  "Procuracy" = lm.procuracy, "United Nations" = lm.un),
  title = "Determinants of trust in institutions",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),

```

```
stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
output = "kableExtra") %>%
kable_styling(latex_options = c("scale_down", "hold_position"))
```

Table 1: Determinants of trust in institutions

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.13* (0.05)	0.05 (0.06)	0.14* (0.06)	0.17** (0.06)	0.07 (0.06)	0.19*** (0.06)	0.21*** (0.06)	0.08 (0.06)	0.05 (0.06)	0.12+ (0.07)
Post-election	0.11 (0.09)	0.19+ (0.10)	0.12 (0.11)	0.01 (0.10)	0.15 (0.11)	0.04 (0.11)	-0.02 (0.11)	0.08 (0.10)	0.18+ (0.11)	-0.11 (0.11)
Protest day	0.22+ (0.13)	0.53*** (0.14)	0.35* (0.14)	0.18 (0.16)	0.30* (0.15)	0.20 (0.16)	0.07 (0.16)	0.03 (0.15)	0.15 (0.15)	0.00 (0.16)
Election day	-0.05 (0.12)	-0.23 (0.16)	0.09 (0.15)	0.11 (0.17)	0.07 (0.15)	-0.19 (0.14)	-0.25+ (0.15)	-0.18 (0.13)	0.03 (0.14)	-0.43** (0.16)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.088	0.066	0.052	0.085	0.057	0.072	0.085	0.062	0.054	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Extending measures of trust

Including the Orthodox Church

```
data <- data %>%
  mutate(trustgovind.withchurch = (trustarmy + trustpolice + trustfsb + trustcourt +
    trustmungov + trustfedgov + trustduma + trustprocuracy +
    trustchurch)/9)

lm.indexwithchurch <- lm(trustgovind.withchurch ~ period3 + period2min + protestday +
  electday + wealth + education + okrug, data = data)

lm.church <- lm(trustchurch ~ period3 + period2min + protestday + electday + wealth +
  education + okrug, data = data)

#reporting in one table
modelsummary(list("Gov Index (with Church)" = lm.indexwithchurch,
  "Church" = lm.church),
  title = "Accounting for the role of Church",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

Leave-one-out indices

```
data <- data %>%
  mutate(trustgovind.noarmy = (trustpolice + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nopolice = (trustarmy + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nofsb = (trustpolice + trustarmy + trustcourt + trustmungov +
```

Table 2: Accounting for the role of Church

	Gov Index (with Church)	Church
Post-protest	0.13** (0.05)	0.10+ (0.06)
Post-election	0.08 (0.09)	0.06 (0.11)
Protest day	0.17 (0.12)	0.03 (0.16)
Election day	-0.07 (0.12)	-0.30 (0.20)
Num.Obs.	1192	1445
R2	0.087	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```

trustfedgov + trustduma + trustprocuracy)/7,
trustgovind.nocourt = (trustpolice + trustfsb + trustarmy + trustmungov +
  trustfedgov + trustduma + trustprocuracy)/7,
trustgovind.nomungov = (trustpolice + trustfsb + trustcourt + trustarmy +
  trustfedgov + trustduma + trustprocuracy)/7,
trustgovind.nofedgov = (trustpolice + trustfsb + trustcourt + trustmungov +
  trustarmy + trustduma + trustprocuracy)/7,
trustgovind.noduma = (trustpolice + trustfsb + trustcourt + trustmungov +
  trustfedgov + trustarmy + trustprocuracy)/7,
trustgovind.noproc = (trustpolice + trustfsb + trustcourt + trustmungov +
  trustfedgov + trustduma + trustarmy)/7)

lm.noarmy <- lm(trustgovind.noarmy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nopolice <- lm(trustgovind.nopolice ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nofsb <- lm(trustgovind.nofsb ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nocourt <- lm(trustgovind.nocourt ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nomungov <- lm(trustgovind.nomungov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nofedgov <- lm(trustgovind.nofedgov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.noduma <- lm(trustgovind.noduma ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.noproc <- lm(trustgovind.noproc ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

```

```
#reporting in one table
modelsummary(list("No Army" = lm.noarmy, "No Police" = lm.nopolice,
  "No FSB" = lm.nofsb, "No Court" = lm.nocourt,
  "No Municipal Gov" = lm.nomungov, "No Federal Gov" = lm.nofedgov,
  "No Duma" = lm.noduma, "No Procuracy" = lm.noproc),
  title = "Robustness check on Gov't Trust: Leave-one-out indices",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

Table 3: Robustness check on Gov't Trust: Leave-one-out indices

	No Army	No Police	No FSB	No Court	No Municipal Gov	No Federal Gov	No Duma	No Procuracy
Post-protest	0.14** (0.05)	0.13** (0.05)	0.11* (0.05)	0.13** (0.05)	0.12* (0.05)	0.12* (0.05)	0.12* (0.05)	0.14** (0.05)
Post-election	0.08 (0.10)	0.09 (0.09)	0.12 (0.08)	0.10 (0.09)	0.13 (0.09)	0.14 (0.09)	0.13 (0.09)	0.08 (0.09)
Protest day	0.16 (0.14)	0.18 (0.13)	0.22+ (0.12)	0.20 (0.13)	0.24+ (0.13)	0.26* (0.13)	0.27* (0.13)	0.21 (0.13)
Election day	-0.03 (0.12)	-0.07 (0.12)	-0.09 (0.12)	-0.07 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.05 (0.12)
Num.Obs.	1242	1235	1332	1243	1235	1236	1242	1260
R2	0.082	0.088	0.078	0.089	0.090	0.089	0.088	0.087

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Sub-indices (based on Schneider 2017)

```
#model 1: fedgov + дума + мунгов
data <- data %>%
  mutate(alt.trust.index1 = (trustfedgov + trustduma + trustmungov)/3)

lm.alt1 <- lm(alt.trust.index1 ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

##NOTE: Schneider distinguishes between police and armed forces, but the role of
#the FSB seems ambiguous within that. for that reason, models 2-5 differentially
#include/exclude the FSB to see if that has any effect on results.

#model 2: fedgov + дума + police + army
data <- data %>%
  mutate(alt.trust.index2 = (trustfedgov + trustduma + trustpolice + trustarmy)/4)

lm.alt2 <- lm(alt.trust.index2 ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

#model 3: fedgov + дума + police + army + fsb
data <- data %>%
  mutate(alt.trust.index3 = (trustfedgov + trustduma + trustpolice + trustfsb +
    trustarmy)/5)
```



```

lm.alt3 <- lm(alt.trust.index3 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#model 4: fedgov + дума + court + police
data <- data %>%
  mutate(alt.trust.index4 = (trustfedgov + trustduma + trustpolice + trustcourt)/4)

lm.alt4 <- lm(alt.trust.index4 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#model 5: fedgov + дума + court + police + fsb
data <- data %>%
  mutate(alt.trust.index5 = (trustfedgov + trustduma + trustpolice + trustcourt +
                             trustfsb)/4)

lm.alt5 <- lm(alt.trust.index5 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#reporting in one table
modelsummary(list("Model 1" = lm.alt1, "Model 2.1" = lm.alt2,
                  "Model 2.2" = lm.alt3,
                  "Model 3.1" = lm.alt4,
                  "Model 3.2" = lm.alt5),
              title = "Robustness check on Gov't Trust: Alternative sub-indices",
              caption = "Sub-indices constructed following Schneider (2017)",
              vcov = "HC2",
              coef_map = c("period3" = "Post-protest",
                           "period2min" = "Post-election", "protestday" = "Protest day",
                           "electday" = "Election day"),
              stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
              add_rows = tribble(
                ~term, ~"Model 1", ~"Model 2.1", ~"Model 2.2", ~"Model 3.1", ~"Model 3.2",
                "Fed Gov", "Yes", "Yes", "Yes", "Yes", "Yes",
                "Duma", "Yes", "Yes", "Yes", "Yes", "Yes",
                "Mun Gov", "Yes", " ", " ", " ", " ",
                "Police", " ", "Yes", "Yes", "Yes", "Yes",
                "Army", " ", "Yes", "Yes", " ", " ",
                "Court", " ", " ", " ", " ", "Yes", "Yes",
                "FSB", " ", " ", " ", "Yes", " ", "Yes"),
              output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))

```

Standardized index

```

data <- data %>%
  mutate(trustarmy.z = as.numeric(scale(trustarmy)),
         trustpolice.z = as.numeric(scale(trustpolice)),
         trustfsb.z = as.numeric(scale(trustfsb)),
         trustcourt.z = as.numeric(scale(trustcourt)),
         trustmungov.z = as.numeric(scale(trustmungov)),
         trustfedgov.z = as.numeric(scale(trustfedgov)),
         trustduma.z = as.numeric(scale(trustduma)),

```

Table 4: Sub-indices constructed following Schneider (2017)

	Model 1	Model 2.1	Model 2.2	Model 3.1	Model 3.2
Post-protest	0.16** (0.05)	0.12* (0.05)	0.14** (0.05)	0.13** (0.05)	0.18** (0.06)
Post-election	0.04 (0.10)	0.11 (0.08)	0.08 (0.09)	0.09 (0.09)	0.08 (0.12)
Protest day	0.12 (0.15)	0.26* (0.12)	0.23+ (0.13)	0.16 (0.13)	0.20 (0.17)
Election day	-0.21 (0.13)	-0.14 (0.12)	-0.06 (0.13)	-0.06 (0.12)	-0.01 (0.15)
Num.Obs.	1438	1417	1282	1388	1275
R2	0.079	0.085	0.094	0.070	0.082
Fed Gov	Yes	Yes	Yes	Yes	Yes
Duma	Yes	Yes	Yes	Yes	Yes
Mun Gov	Yes				
Police		Yes	Yes	Yes	Yes
Army		Yes	Yes		
Court				Yes	Yes
FSB			Yes		Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
trustprocuracy.z = as.numeric(scale(trustprocuracy)),
trustgovindex.standardized = (trustarmy.z + trustpolice.z + trustfsb.z +
                              trustcourt.z + trustmungov.z + trustfedgov.z +
                              trustduma.z + trustprocuracy.z)/8)

lm.standardized <- lm(trustgovindex.standardized ~ period3 + period2min + protestday +
                      electday + wealth + education + okrug, data = data)

modelsummary(lm.standardized,
  title = "Robustness check on Gov't Trust: Index with Z-Scored Items",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

Factor-score index

```
#Factor loading analysis
fa.analysis.data <- data %>%
  select(trustarmy, trustpolice, trustfsb, trustcourt, trustmungov,
    trustfedgov, trustduma, trustprocuracy)

fa.fit <- fa(fa.analysis.data, nfactors = 1, rotate = "oblimin", fm = "ml")

fa.parallel(fa.analysis.data, fm = "ml")
```

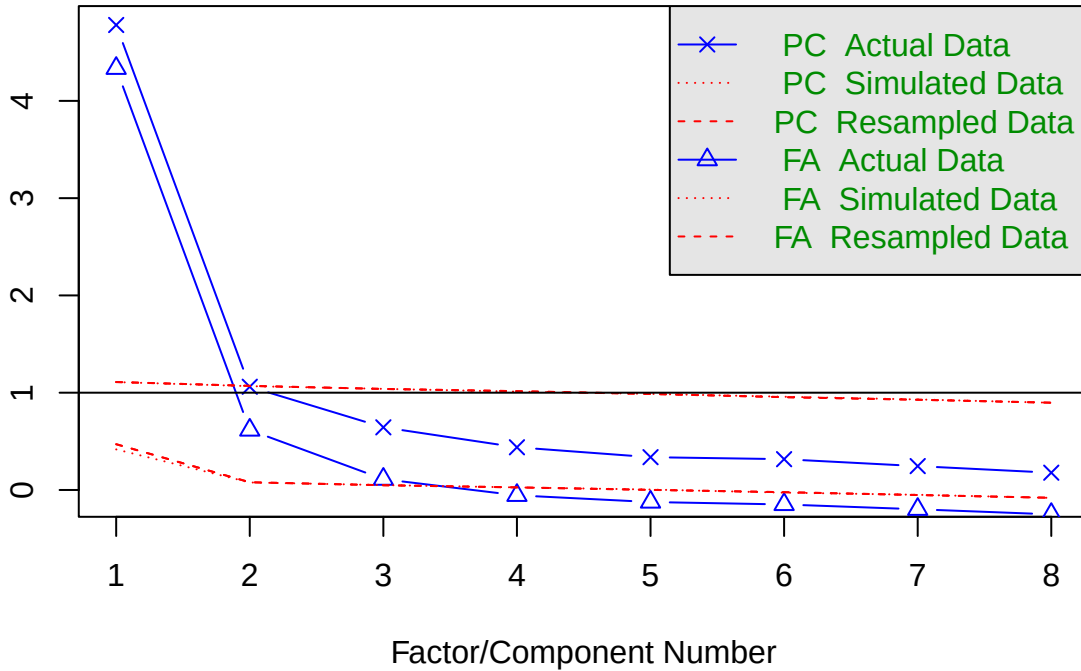
Table 5: Robustness check on Gov't Trust: Index with Z-Scored Items

	(1)
Post-protest	0.13*
	(0.05)
Post-election	0.11
	(0.09)
Protest day	0.22+
	(0.13)
Election day	-0.05
	(0.12)
Num.Obs.	1232
R2	0.087

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

eigenvalues of principal components and factor analysis

Parallel Analysis Scree Plots



Paral-

lel analysis suggests that the number of factors = 3 and the number of components = 1

```
cr.alpha.data <- alpha(fa.analysis.data)
#crophback's alpha ~ 0.903
data.frame(raw_alpha = cr.alpha.data$total$raw_alpha,
           std.alpha = cr.alpha.data$total$std.alpha)
```

```
raw_alpha std.alpha 1 0.903453 0.9027666
```

```
#Factor-score-weighted index
fa.fit$loadings
```

Loadings: ML1

trustarmy 0.530 trustpolice 0.691 trustfsb 0.703 trustcourt 0.680 trustmungov 0.820 trustfedgov 0.846
trustduma 0.817 trustprocuracy 0.718

ML1

SS loadings 4.284 Proportion Var 0.536

```
weights <- as.numeric(fa.fit$loadings)
weights.norm <- weights/sum(weights)
weights.norm
```

[1] 0.09125392 0.11901039 0.12114956 0.11710733 0.14129712 0.14580924 0.14074087 [8] 0.12363157

```
data <- data %>%
  mutate(trustgovind.factor.weight = (0.091*trustarmy) + (0.119*trustpolice) +
    (0.121*trustfsb) + (0.117*trustcourt) + (0.141*trustmungov) +
    (0.146*trustfedgov) + (0.141*trustduma) + (0.124*trustprocuracy))
#all weights add up to 1

lm.factor.weighted <- lm(trustgovind.factor.weight ~ period3 + period2min + protestday +
  electday + wealth + education + okrug, data = data)

modelsummary(lm.factor.weighted,
  title = "Robustness check on Gov't Trust: Factor Loading Score-Weighted Index",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

Table 6: Robustness check on Gov't Trust: Factor Loading Score-Weighted Index

	(1)
Post-protest	0.13** (0.05)
Post-election	0.09 (0.09)
Protest day	0.19 (0.13)
Election day	-0.05 (0.12)
Num.Obs.	1232
R2	0.086

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
#additional confirmatory factor analysis
model <- "trust =~ trustarmy + trustpolice + trustfsb + trustcourt +
  trustmungov + trustfedgov + trustduma + trustprocuracy"

model.fit <- cfa(model, data = data)
summary(model.fit, fit.measures = TRUE)
```

```

## lavaan 0.6-21 ended normally after 25 iterations
##
## Estimator ML
## Optimization method NLMINB
## Number of model parameters 16
##
## Used Total
## Number of observations 1249 1550
##
## Model Test User Model:
##
## Test statistic 1194.227
## Degrees of freedom 20
## P-value (Chi-square) 0.000
##
## Model Test Baseline Model:
##
## Test statistic 6609.970
## Degrees of freedom 28
## P-value 0.000
##
## User Model versus Baseline Model:
##
## Comparative Fit Index (CFI) 0.822
## Tucker-Lewis Index (TLI) 0.750
##
## Loglikelihood and Information Criteria:
##
## Loglikelihood user model (H0) -11465.307
## Loglikelihood unrestricted model (H1) NA
##
## Akaike (AIC) 22962.614
## Bayesian (BIC) 23044.696
## Sample-size adjusted Bayesian (SABIC) 22993.873
##
## Root Mean Square Error of Approximation:
##
## RMSEA 0.217
## 90 Percent confidence interval - lower 0.206
## 90 Percent confidence interval - upper 0.227
## P-value H_0: RMSEA <= 0.050 0.000
## P-value H_0: RMSEA >= 0.080 1.000
##
## Standardized Root Mean Square Residual:
##
## SRMR 0.090
##
## Parameter Estimates:
##
## Standard errors Standard
## Information Expected
## Information saturated (h1) model Structured
##
## Latent Variables:

```

```
##               Estimate Std.Err z-value P(>|z|)
## trust =~
##   trustarmy      1.000
##   trustpolice    1.235    0.070   17.707    0.000
##   trustfsb       1.278    0.073   17.624    0.000
##   trustcourt     1.239    0.070   17.722    0.000
##   trustmungov    1.513    0.077   19.563    0.000
##   trustfedgov    1.624    0.082   19.869    0.000
##   trustduma      1.552    0.079   19.555    0.000
##   trustprocuracy 1.299    0.072   18.145    0.000
##
## Variances:
##               Estimate Std.Err z-value P(>|z|)
##   .trustarmy      0.772    0.032   24.198    0.000
##   .trustpolice    0.494    0.021   23.092    0.000
##   .trustfsb       0.546    0.024   23.152    0.000
##   .trustcourt     0.494    0.021   23.081    0.000
##   .trustmungov    0.281    0.014   19.913    0.000
##   .trustfedgov    0.248    0.014   18.360    0.000
##   .trustduma      0.297    0.015   19.946    0.000
##   .trustprocuracy 0.457    0.020   22.718    0.000
##   trust           0.300    0.031    9.732    0.000
```

##Comparative fit measurements - closer to 1 the better

#CFI: 0.822

#TLI: 0.75

#RMSEA: 0.217 (penalized model complexity)

#SRMR: 0.09 (difference between observed and model-implied correlations)

#NOTE: Borderline/poorer statistics more common with survey data (sensitive to N).

Mechanisms

Figure 1

```
data <- data %>%
  mutate(period123 = case_when(period1min == 1 ~ 1,
                                period2min == 1 ~ 2,
                                period3 == 1 ~ 3,
                                TRUE ~ NA_real_),
    opposition = yab + pat + sr + rc,
    party = case_when(ur == 1 ~ 1,
                      comp == 1 ~ 2,
                      ldpr == 1 ~ 3,
                      opposition == 1 ~ 4,
                      nopart == 1 ~ 5,
                      dk == 1 ~ 6,
                      TRUE ~ 0),
    partynames = case_when(party == 1 ~ "UR",
                           party == 2 ~ "CPRF",
                           party == 3 ~ "LDPR",
                           party == 4 ~ "Opposition",
                           party == 5 ~ "Non-voters",
```

```

    party == 6 ~ "Undecided",
    TRUE ~ NA_character_),
  partynames = factor(partynames)) %>%
mutate(trustgovind1 = ifelse(period1min == 1, 1, 0),
  trustgovind2 = ifelse(period2min == 1, 1, 0),
  trustgovind3 = ifelse(period3 == 1, 1, 0))

data %>%
  group_by(partynames, period123) %>%
  summarize(mean.trust = mean(trustgovind, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(x = partynames,
    y = mean.trust,
    fill = factor(period123))) +
  geom_col(position = "dodge") +
  labs(y = "Trust in Government Index", fill = "Period")

```

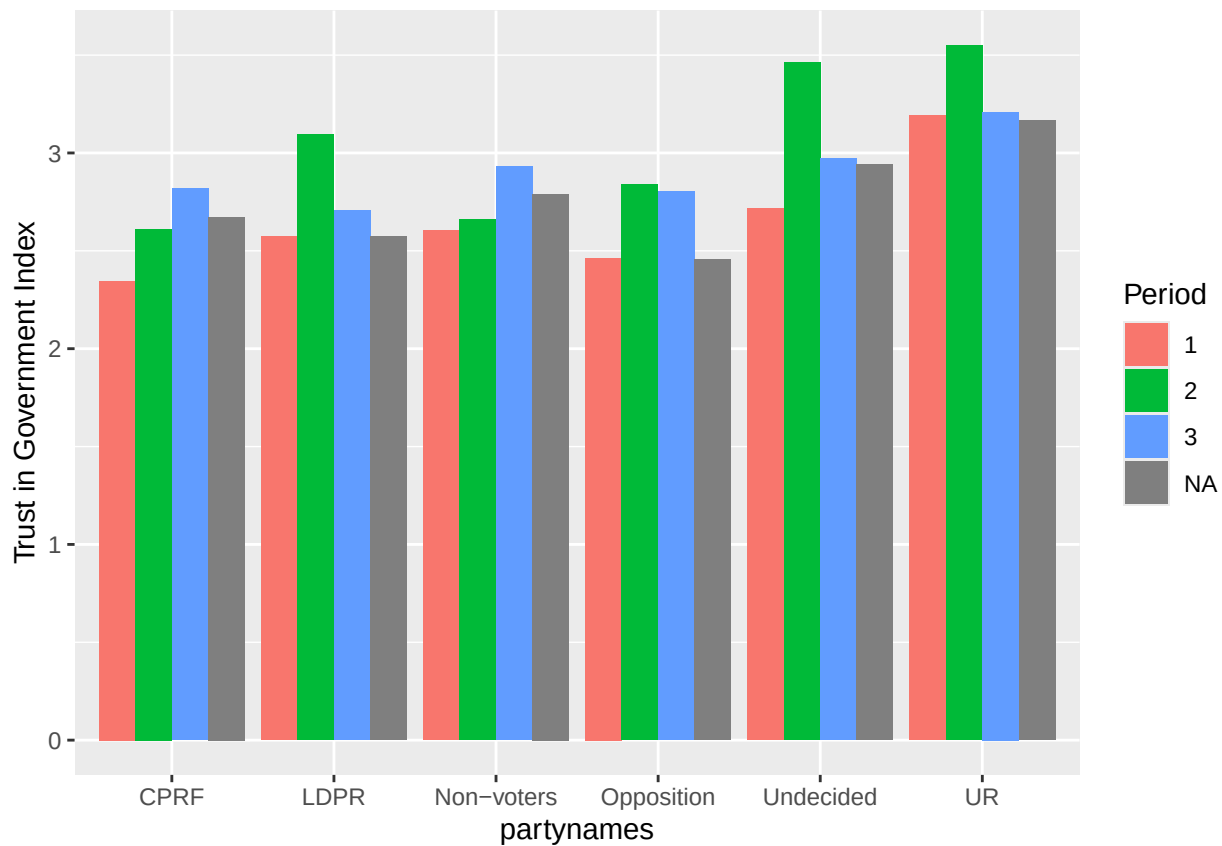


Table 3

```

data <- data %>%
  mutate(urafterprotest = ur*period3,
    urpostelection = ur*period2min,
    apolitical = nopart + dk,
    apolafterprotest = apolitical*period3,
    apolpostelection = apolitical*period2min)

lm.interact.govindex <- lm(trustgovind ~ period3 + period2min + protestday + electday +

```

```

        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.army <- lm(trustarmy ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.police <- lm(trustpolice ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.fsb <- lm(trustfsb ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.court <- lm(trustcourt ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.mungov <- lm(trustmungov ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.fedgov <- lm(trustfedgov ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.duma <- lm(trustduma ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.procuracy <- lm(trustprocuracy ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

lm.interact.un <- lm(trustun ~ period3 + period2min + protestday + electday +
        ur + urafterprotest + urpostelection +
        apolitical + apolafterprotest + apolpostelection +
        wealth + education + okrug, data = data)

modelsummary(list("Gov Index" = lm.interact.govindex, "Army" = lm.interact.army, "Police" = lm.interact.police,
        "FSB" = lm.interact.fsb, "Court" = lm.interact.court, "Municipal Gov" = lm.interact.mungov,
        "Federal Gov" = lm.interact.fedgov, "Duma" = lm.interact.duma,
        "Procuracy" = lm.interact.procuracy, "United Nations" = lm.interact.un),

```



```

title = "Determinants of trust in institutions",
vcov = "HC2",
coef_map = c("period3" = "Post-protest",
"period2min" = "Post-election", "protestday" = "Protest day",
"electday" = "Election day", "ur" = "United Russia", "urafterprotest" = "United Russia x P
stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
output = "kableExtra") %>%
kable_styling(latex_options = c("scale_down", "hold_position"))

```

Table 7: Determinants of trust in institutions

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.18* (0.08)	0.08 (0.09)	0.08 (0.09)	0.17+ (0.10)	0.11 (0.09)	0.28** (0.09)	0.26** (0.09)	0.21* (0.09)	0.13 (0.09)	0.08 (0.10)
Post-election	0.09 (0.13)	0.25* (0.13)	0.16 (0.14)	-0.01 (0.16)	0.16 (0.16)	-0.09 (0.15)	-0.13 (0.14)	-0.01 (0.14)	0.06 (0.16)	-0.11 (0.17)
Protest day	0.22+ (0.13)	0.52*** (0.14)	0.35* (0.14)	0.19 (0.16)	0.28+ (0.15)	0.22 (0.16)	0.09 (0.16)	0.03 (0.15)	0.14 (0.16)	0.02 (0.17)
Election day	-0.01 (0.11)	-0.20 (0.15)	0.13 (0.15)	0.15 (0.16)	0.10 (0.15)	-0.14 (0.15)	-0.20 (0.15)	-0.12 (0.14)	0.08 (0.14)	-0.43** (0.16)
United Russia	0.59*** (0.09)	0.42*** (0.13)	0.58*** (0.11)	0.58*** (0.12)	0.40*** (0.12)	0.66*** (0.11)	0.77*** (0.11)	0.74*** (0.12)	0.49*** (0.12)	0.13 (0.14)
United Russia x Post-Protest	-0.14 (0.14)	-0.07 (0.17)	-0.12 (0.16)	-0.14 (0.18)	-0.19 (0.17)	-0.17 (0.16)	-0.19 (0.16)	-0.34* (0.17)	-0.24 (0.18)	-0.07 (0.20)
United Russia x Post-Election	0.12 (0.20)	0.09 (0.21)	0.15 (0.23)	0.21 (0.25)	0.24 (0.21)	0.25 (0.21)	0.27 (0.21)	0.30 (0.22)	0.30 (0.24)	0.17 (0.26)
Apolitical	0.17* (0.07)	0.16+ (0.09)	0.09 (0.08)	0.12 (0.09)	0.16+ (0.09)	0.15+ (0.08)	0.15+ (0.08)	0.20* (0.08)	0.21* (0.08)	-0.08 (0.10)
Apolitical x Post-Protest	-0.05 (0.10)	-0.04 (0.12)	0.16 (0.11)	0.06 (0.12)	-0.02 (0.11)	-0.14 (0.11)	-0.06 (0.12)	-0.18 (0.11)	-0.10 (0.11)	0.12 (0.14)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.134	0.082	0.084	0.111	0.070	0.109	0.132	0.099	0.070	0.044

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 2

```

coefs.robust <- coeftest(lm.interact.govindex, vcov = vcovHC(lm.interact.govindex, type = "HC2"))

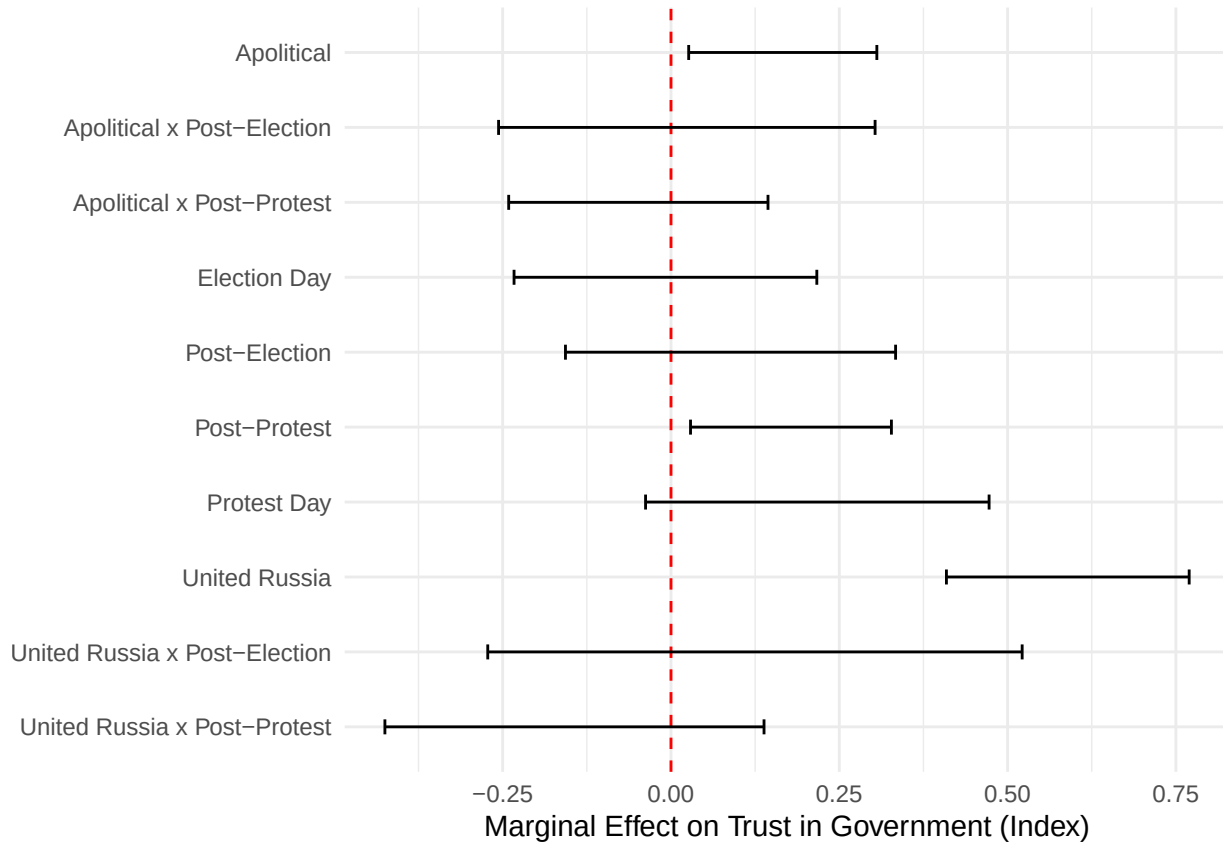
vars <- c("period3", "period2min", "protestday", "electday", "ur", "urafterprotest", "urpostelection", "apolitical", "apolafterprotest", "apolpostelection")
var.names <- c("period3" = "Post-Protest",
"period2min" = "Post-Election",
"protestday" = "Protest Day",
"electday" = "Election Day",
"ur" = "United Russia",
"urafterprotest" = "United Russia x Post-Protest",
"urpostelection" = "United Russia x Post-Election",
"apolitical" = "Apolitical",
"apolafterprotest" = "Apolitical x Post-Protest",
"apolpostelection" = "Apolitical x Post-Election")

margeffects.data <- data.frame(vars = vars,
beta = coefs.robust[vars, "Estimate"],
se = coefs.robust[vars, "Std. Error"])

margeffects.data <- margeffects.data %>%
mutate(ci.lower = beta - 1.96 * se,
ci.upper = beta + 1.96 * se,
var.label = factor(var.names[vars]))

```

```
ggplot(margeffects.data, aes(x = beta, y = var.label)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  geom_errorbar(aes(xmin = ci.lower, xmax = ci.upper), height = 0.2) +
  scale_y_discrete(limits = rev) +
  labs(x = "Marginal Effect on Trust in Government (Index)", y = NULL) +
  theme_minimal()
```



Appendix

Appendix I: Distribution of respondents across districts by periods

Appendix II: Balance tests

Appendix III: Differences in partisanship across periods

Appendix IV: Non-responses

Appendix V: Daily responses